

Lecture 20: Two-Stage Least Squares

Economics 326 — Introduction to Econometrics II

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Multiple linear IV model

- In practice, we often estimate models with multiple endogenous and exogenous regressors.
- Example:

$$\ln \text{Wage}_i = \gamma_0 + \gamma_1 \text{Age}_i + \gamma_2 \text{Sex}_i + \beta_1 \text{Educ}_i + \beta_2 \text{Children}_i + U_i.$$

- **Exogenous** regressors: age, sex, and a constant.
- **Endogenous** regressors: education and children (family size).

Multiple linear IV model

- General model:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i,$$

where

- y_i is the dependent variable.
- γ_0 is the intercept: $E[U_i] = 0$.
- $X_{1,i}, \dots, X_{k,i}$ are k **exogenous** regressors:

$$\text{Cov}(X_{1,i}, U_i) = \dots = \text{Cov}(X_{k,i}, U_i) = 0.$$

- $Y_{1,i}, \dots, Y_{m,i}$ are m **endogenous** regressors:

$$\text{Cov}(Y_{1,i}, U_i) \neq 0, \dots, \text{Cov}(Y_{m,i}, U_i) \neq 0.$$

Identification problem

- There are $k + 1 + m$ unknown coefficients:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i.$$

- The exogeneity conditions give only $k + 1$ equations:

$$E[U_i] = 0, \quad \text{Cov}(X_{j,i}, U_i) = 0, \quad j = 1, \dots, k.$$

- More unknowns than equations: the covariances between X 's, Y 's, and y do not suffice to recover the coefficients.
- Without additional information, the coefficients are **not identified** even at the population level.
- We need **at least** m additional equations!

Instrumental variables

- Suppose we observe l additional **exogenous** variables (IVs) $Z_{1,i}, \dots, Z_{l,i}$.
- The IVs are **excluded** from the **structural equation**:

$$y_i = \gamma_0 + \gamma_1 X_{1,i} + \dots + \gamma_k X_{k,i} + \beta_1 Y_{1,i} + \dots + \beta_m Y_{m,i} + U_i,$$

so there are still $k + 1 + m$ structural coefficients.

- Since the IVs are exogenous, we gain l additional moment conditions:

$$\text{Cov}(Z_{j,i}, U_i) = 0, \quad j = 1, \dots, l,$$

bringing the total to $k + 1 + l$ equations.

- **Necessary condition** for identification: $l \geq m$.

First-stage equations

- The IVs must also determine the endogenous regressors.
- The system consists of the **structural equation** and m **first-stage (reduced-form)** equations:

$$\begin{aligned} Y_{1,i} &= \pi_{0,1} + \pi_{1,1} Z_{1,i} + \dots + \pi_{l,1} Z_{l,i} \\ &\quad + \pi_{l+1,1} X_{1,i} + \dots + \pi_{l+k,1} X_{k,i} + V_{1,i}, \\ &\vdots \\ Y_{m,i} &= \pi_{0,m} + \pi_{1,m} Z_{1,i} + \dots + \pi_{l,m} Z_{l,i} \\ &\quad + \pi_{l+1,m} X_{1,i} + \dots + \pi_{l+k,m} X_{k,i} + V_{m,i}. \end{aligned}$$

- The exogenous regressors X 's appear in the first-stage equations because they can be correlated with Y 's.
- The X 's and Z 's are **uncorrelated** with all errors U and V 's.

Order condition for identification

- Necessary condition: for every endogenous regressor Y , there must be **at least** one excluded exogenous variable Z :

$$l \geq m.$$

- $l = m$: the system is **exactly identified**.
- $l > m$: the system is **overidentified**.
- $l < m$: the system is **underidentified**; the structural coefficients γ 's and β 's cannot be estimated.

2SLS: the first stage

- The first-stage equations:

$$\begin{aligned} Y_{1,i} &= \pi_{0,1} + \pi_{1,1} Z_{1,i} + \dots + \pi_{l,1} Z_{l,i} \\ &\quad + \pi_{l+1,1} X_{1,i} + \dots + \pi_{l+k,1} X_{k,i} + V_{1,i}, \\ &\vdots \\ Y_{m,i} &= \pi_{0,m} + \pi_{1,m} Z_{1,i} + \dots + \pi_{l,m} Z_{l,i} \\ &\quad + \pi_{l+1,m} X_{1,i} + \dots + \pi_{l+k,m} X_{k,i} + V_{m,i}. \end{aligned}$$

- All right-hand side variables are exogenous.
- The first-stage coefficients π 's can be estimated consistently by OLS, regressing each Y against Z 's and X 's.

2SLS: the first stage

- Let $\hat{\pi}$'s denote the OLS estimators of π 's. Obtain the fitted values:

$$\begin{aligned}\hat{Y}_{1,i} &= \hat{\pi}_{0,1} + \hat{\pi}_{1,1}Z_{1,i} + \dots + \hat{\pi}_{l,1}Z_{l,i} \\ &\quad + \hat{\pi}_{l+1,1}X_{1,i} + \dots + \hat{\pi}_{l+k,1}X_{k,i}, \\ &\vdots \\ \hat{Y}_{m,i} &= \hat{\pi}_{0,m} + \hat{\pi}_{1,m}Z_{1,i} + \dots + \hat{\pi}_{l,m}Z_{l,i} \\ &\quad + \hat{\pi}_{l+1,m}X_{1,i} + \dots + \hat{\pi}_{l+k,m}X_{k,i}.\end{aligned}$$

- The $\hat{Y}$'s are functions of Z 's and X 's (all exogenous), so they are asymptotically uncorrelated with the errors.

2SLS: the second stage

- In the second stage, regress (OLS) y against a constant, X 's, and the fitted values $\hat{Y}$'s:

$$\begin{aligned}y_i &= \hat{\gamma}_0^{2SLS} + \hat{\gamma}_1^{2SLS}X_{1,i} + \dots + \hat{\gamma}_k^{2SLS}X_{k,i} \\ &\quad + \hat{\beta}_1^{2SLS}\hat{Y}_{1,i} + \dots + \hat{\beta}_m^{2SLS}\hat{Y}_{m,i} + \hat{U}_i.\end{aligned}$$

- The 2SLS estimators $\hat{\gamma}_0^{2SLS}, \dots, \hat{\gamma}_k^{2SLS}, \hat{\beta}_1^{2SLS}, \dots, \hat{\beta}_m^{2SLS}$ are **consistent** and **asymptotically normal**.
- Standard errors from naïve second-stage OLS are **incorrect**: they do not account for the estimation error in $\hat{\pi}$'s from the first stage.
- Statistical packages report the corrected standard errors.

Example: returns to education

- Estimate the returns to education using the MROZ dataset (Wooldridge, 2006):

$$\ln \text{Wage}_i = \gamma_0 + \beta_1 \text{Educ}_i + \gamma_1 \text{Exper}_i + \gamma_2 \text{Exper}_i^2 + U_i.$$

- **Endogenous**: Educ (education).
- **Instruments**: MotherEduc and FatherEduc (parents' education).
- **Exogenous**: Exper and Exper² (experience).
- The system is **overidentified** ($l = 2 > m = 1$).
- In Stata, use `ivregress 2sls` with options `robust` (heteroskedasticity-robust SEs) and `first` (report the first stage).

Example: returns to education

First-stage regression (Educ on instruments and exogenous regressors):

```
Number of obs   =      428
F(   4,   423)  =      25.76
Prob > F        =      0.0000
R-squared       =      0.2115
Adj R-squared   =      0.2040
Root MSE       =      2.0390
```

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
educ							
exper		.0452254	.0419107	1.08	0.281	-.0371538	.1276046
expersq		-.0010091	.0013233	-0.76	0.446	-.0036101	.0015919

motheduc		.157597	.0354502	4.45	0.000	.0879165	.2272776
fatheduc		.1895484	.0324419	5.84	0.000	.125781	.2533159
_cons		9.10264	.4241444	21.46	0.000	8.268947	9.936333

Both instruments (motheduc, fatheduc) are statistically significant.

Example: returns to education

2SLS second stage:

Instrumental variables (2SLS) regression	Number of obs =	428
	Wald chi2(3) =	18.61
	Prob > chi2 =	0.0003
	R-squared =	0.1357
	Root MSE =	.67155

lwage	Coef.	Robust Std. Err.	z	P> z	[95% Conf. Interval]
educ	.0613966	.0331824	1.85	0.064	-.0036397 .126433
exper	.0441704	.0154736	2.85	0.004	.0138428 .074498
expersq	-.000899	.0004281	-2.10	0.036	-.001738 -.00006
_cons	.0481003	.4277846	0.11	0.910	-.7903421 .8865427

Instrumented: educ

Instruments: exper expersq motheduc fatheduc

OLS for comparison:

lwage	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
educ	.1074896	.013219	8.13	0.000	.0815068 .1334725
exper	.0415665	.015273	2.72	0.007	.0115462 .0715868
expersq	-.0008112	.0004201	-1.93	0.054	-.0016369 .0000145
_cons	-.5220406	.2016505	-2.59	0.010	-.9183996 -.1256815

The 2SLS estimate of the return to education ($\hat{\beta}_1^{2SLS} = 0.061$) is smaller than the OLS estimate ($\hat{\beta}_1^{OLS} = 0.107$), consistent with upward ability bias.